

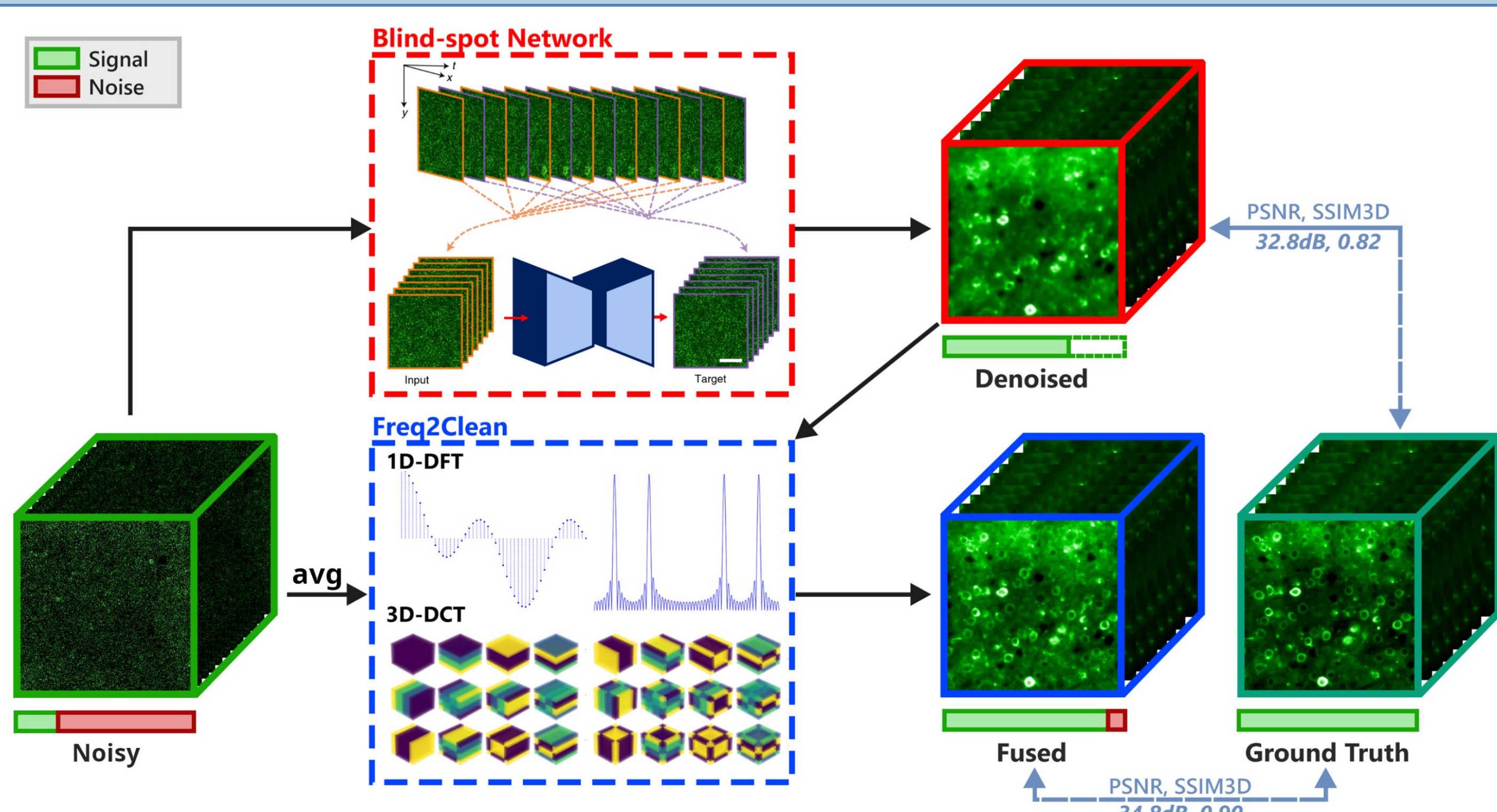


Motivations

- Calcium imaging is a cornerstone of neuroscience, but the **low signal-to-noise ratio (SNR)** of the acquired footage obscures the neural signals of interest.
- Self-supervised denoisers** removed the need for clean ground truth, yet they **oversmooth fine spatial detail** under extreme noise, degrading downstream ROI segmentation and neuron extraction.
- Temporal averaging** raises spatial SNR by a factor of $\sqrt{N}$, but acts as a low-pass filter that **suppresses fast calcium transients**.
- The two signals are **complementary**: averaging wins at low temporal frequencies, the denoiser at high ones.

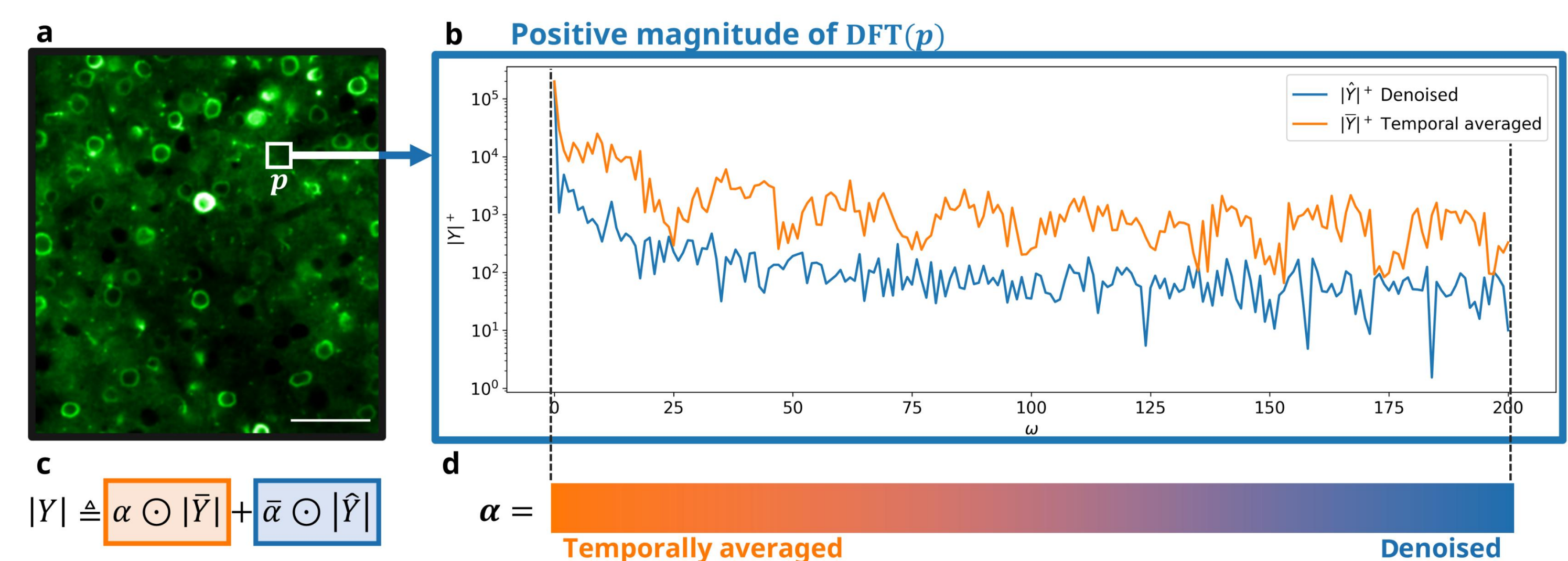
Research Question: Can we restore the fine spatial detail lost by self-supervised denoisers without altering the temporal dynamics of neuronal activity?

PROPOSED APPROACH



- 1D DFT along time:** one N-point DFT per pixel sequence; Hermitian symmetry means only $\lfloor N/2 \rfloor + 1$ bins are needed, halving memory and compute.
- Learned fusion profile:** a coefficient tensor over the 1,501 bins ($N = 3,000$) is optimized once on synthetic NAOMi sequences, then **frozen**.
- Calibration objective:** $L_1 + L_2$ spatial reconstruction, plus a Pearson-correlation temporal-fidelity term and a regulariser discouraging reliance on the averaged signal at high frequencies.
- Resulting profile:** the averaged signal dominates at DC and low frequencies, the denoiser at high frequencies.
- Scalability:** an FFT implementation processes videos in non-overlapping temporal patches of $N = 3,000$ frames (grid-searched trade-off between memory and spectral leakage) and can be batched arbitrarily along the two spatial dimensions, leading to ≈ 5 s for a (6000, 490, 490) video on an RTX 5090.

- Complementary by construction:** under a Poisson–Gaussian model, averaging N frames scales SNR by $\sqrt{N}$, but low-pass filters the video at $1/N$ of the frame rate; the denoiser keeps fast dynamics yet blurs structure.
- Fuse in the frequency domain:** Freq2Clean takes the temporally averaged noisy measurement and the denoiser's prediction and combines them frequency by frequency.
- Magnitude-only fusion:** a convex combination of the two Fourier magnitudes, **retaining the phase of the denoiser** so temporal alignment is untouched.
- Drop-in module:** the baseline denoiser is never modified or retrained.



RESULTS & CONCLUSIONS

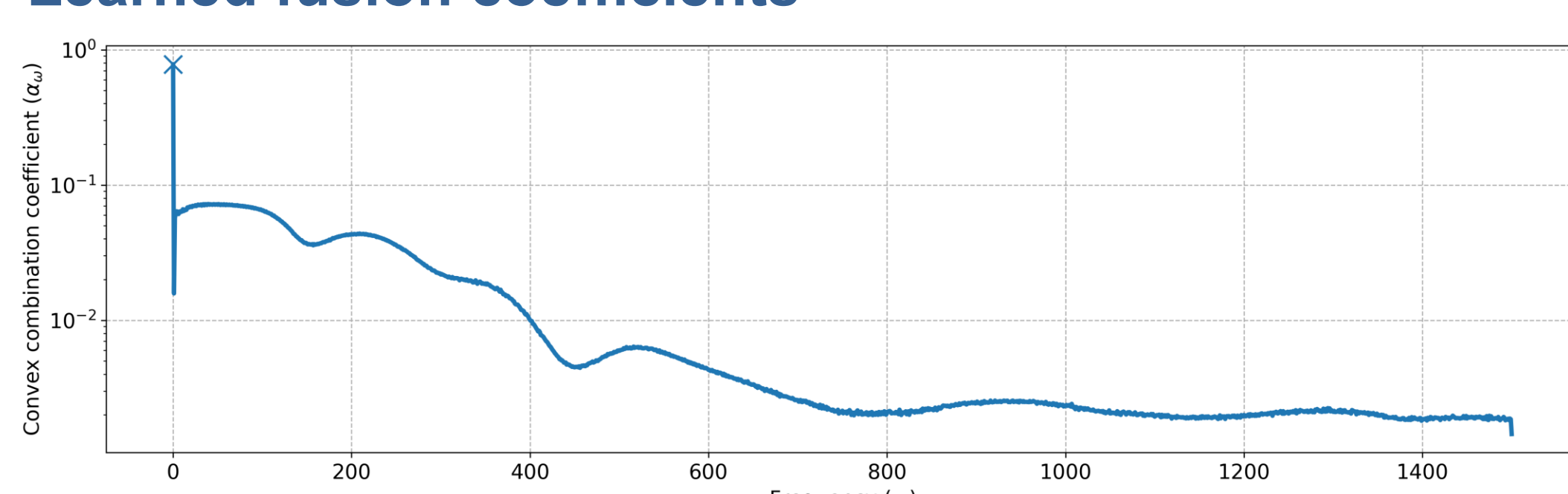
Synthetic NAOMi, SNR – 5.52 dB

	Baseline		+ Freq2Clean	
Denoiser	PSNR _{3D} ↑	SSIM _{3D} ↑	PSNR _{3D} ↑	SSIM _{3D} ↑
BM3D	13.52	0.207	13.74	0.280
BM4D	14.61	0.385	14.79	0.486
Noise2Void	16.35	0.267	17.21	0.288
Noise2Noise	18.64	0.499	19.13	0.594
DeepCAD-RT	27.94	0.760	30.04	0.880
SRDTrans	25.48	0.635	25.57	0.658
DeepVIDv2	20.30	0.455	21.19	0.486
TeD	22.64	0.546	23.22	0.597
FAST	20.91	0.362	22.19	0.495

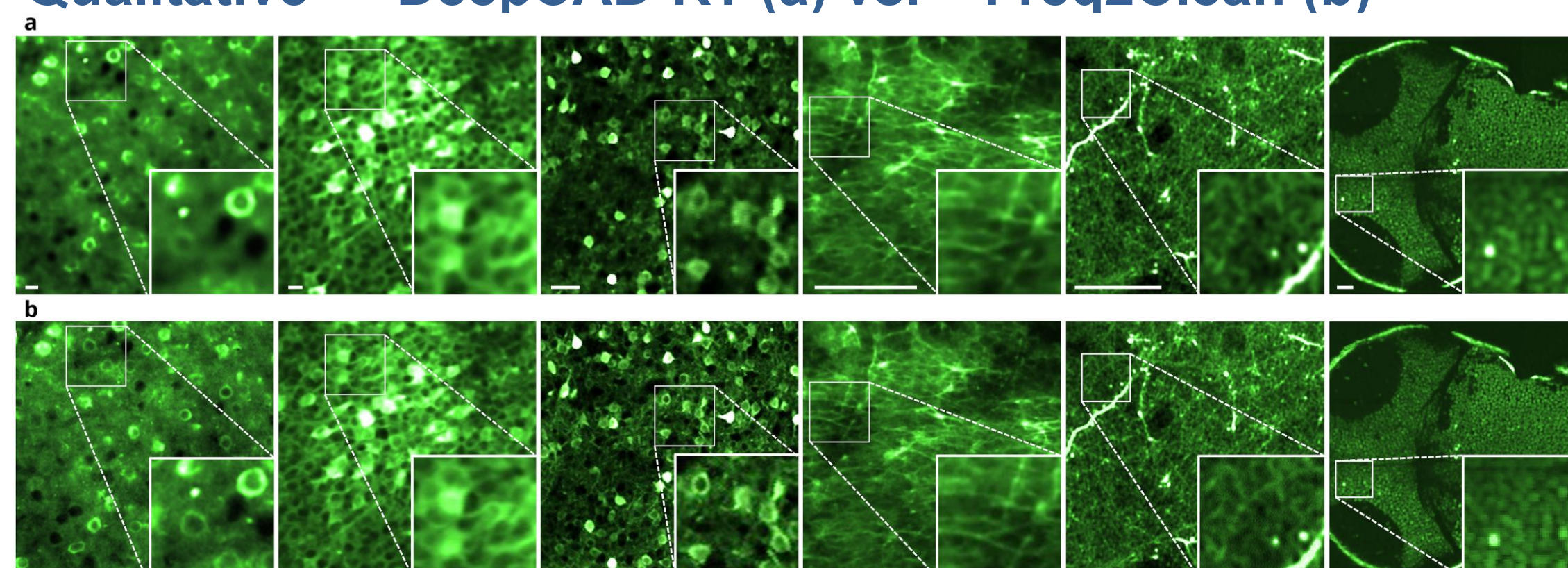
Real recordings, vs. higher-SNR references

	DeepCAD-RT		+ Freq2Clean	
Dataset	PSNR _{3D} ↑	SSIM _{3D} ↑	PSNR _{3D} ↑	SSIM _{3D} ↑
Mouse neuronal pop.	19.33	0.210	19.52	0.244
Zebrafish brain	16.84	0.259	16.87	0.289
Mouse spines (50 mW)	13.38	0.090	13.46	0.092
Mouse spines (115 mW)	13.40	0.149	13.43	0.155

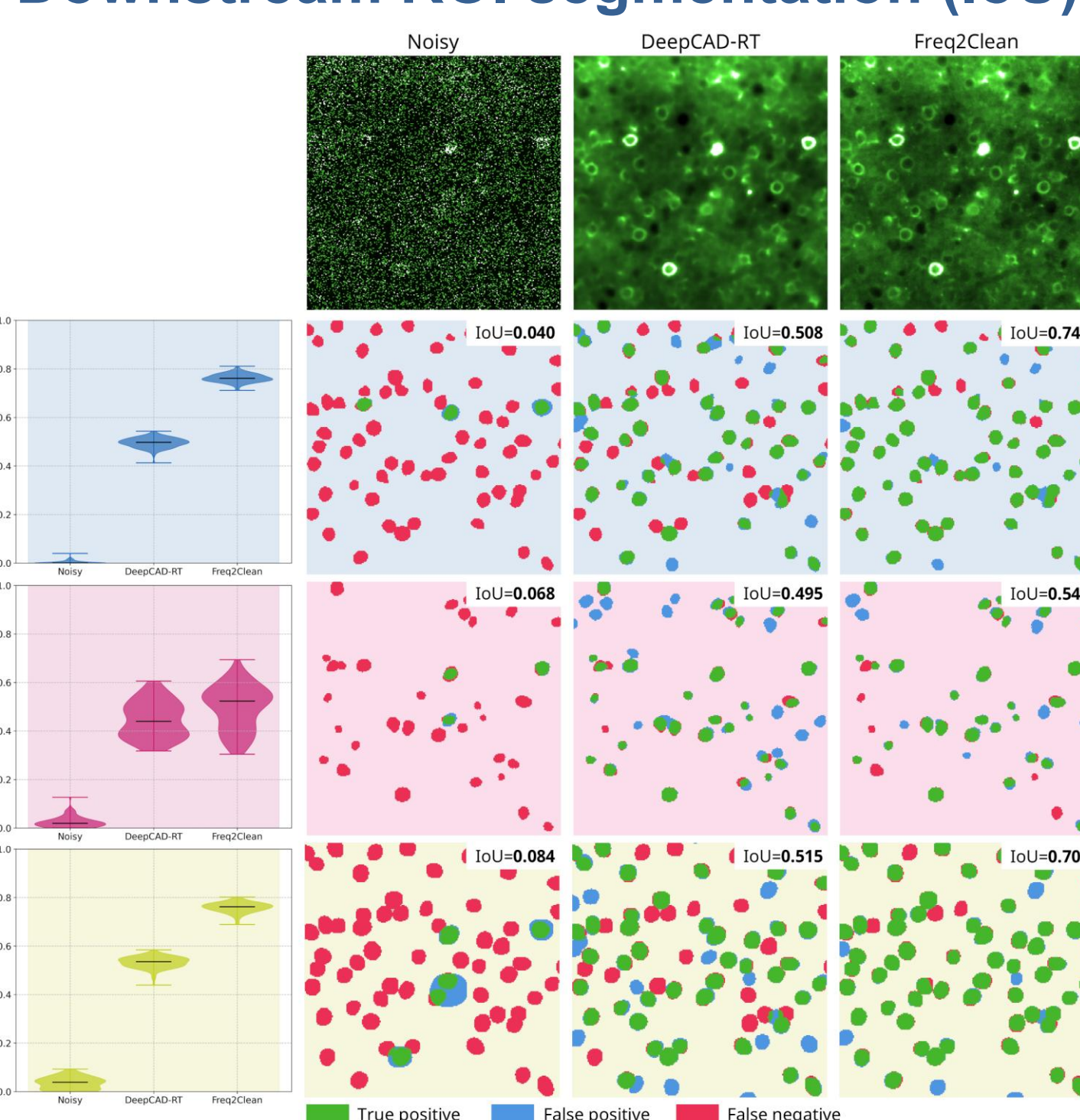
Learned fusion coefficients



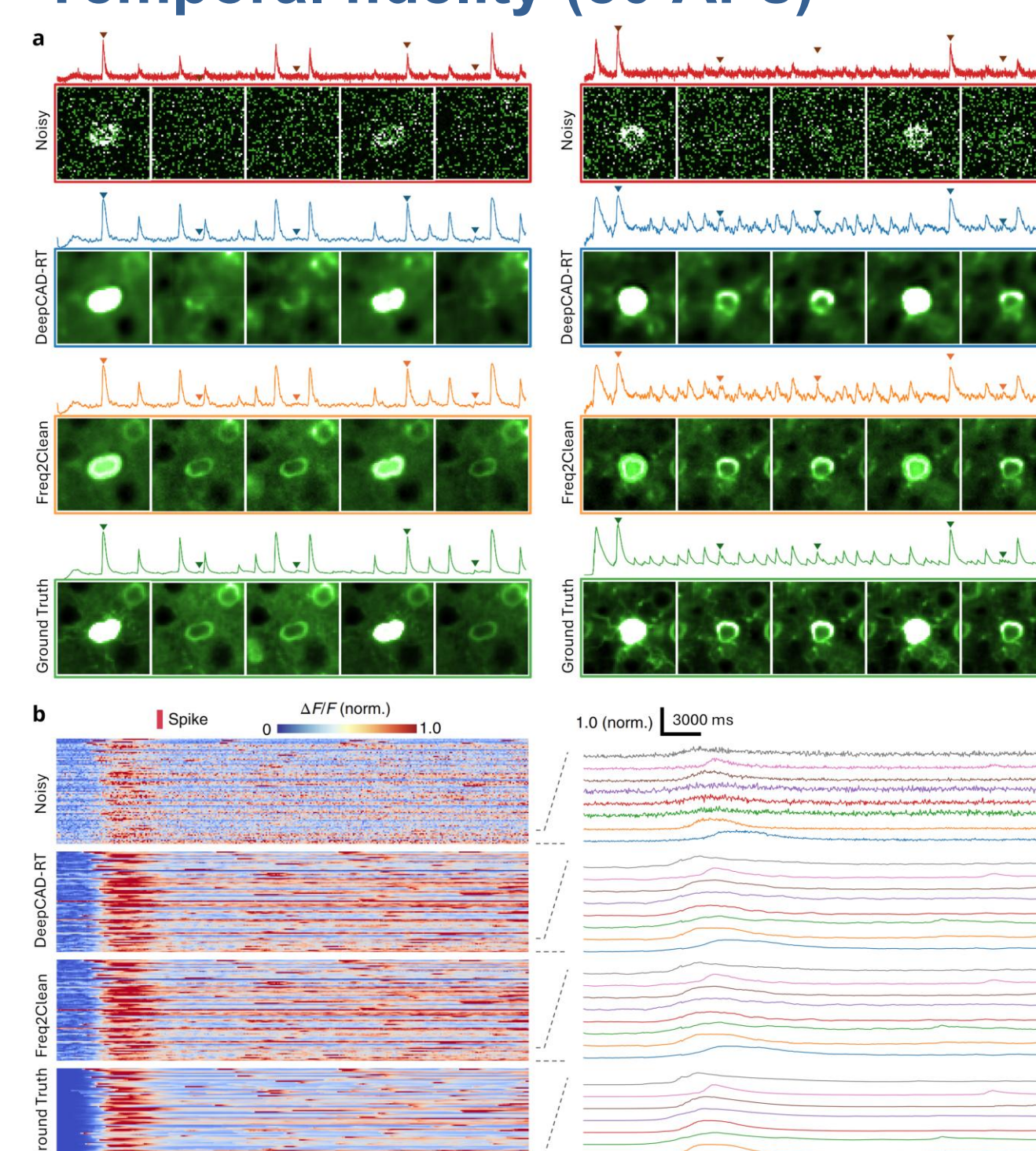
Qualitative — DeepCAD-RT (a) vs. + Freq2Clean (b)



Downstream ROI segmentation (IoU)



Temporal fidelity (80 APs)



Key Findings

- Freq2Clean improves **every denoiser tested** on NAOMi at SNR – 5.52 dB.
- Largest gain on DeepCAD-RT: **+2.10 dB PSNR** (27.94 → 30.04) and **+0.12 SSIM** (0.760 → 0.880).
- Gains **transfer to real recordings**: mouse populations, zebrafish brain and dendritic spines.

- Cellpose, StarDist and CellSAM masks move **closer to clean-data masks**, with lower variance across samples.
- Calcium transients are **unchanged**: temporal dynamics of the baseline denoiser are preserved.
- Ablations: frequency-domain fusion beats **time-domain average blending**, and **1D-DFT beats 3D-DCT** across all denoisers.

- Freq2Clean is a simple, lightweight, **denoiser-agnostic** module that recovers spatial detail lost to extreme noise while leaving temporal dynamics intact.
- It is most effective on still recordings with limited spatial dynamics and very low SNR; requiring multiple consecutive frames currently limits real-time use.
- Future Work:** higher-dimensional data (4D fMRI), complex-valued measurements (frequency-domain FLIM), and end-to-end joint training with state-of-the-art denoisers.